Practical Work (TP3): Light Intensity Acquisition System

Equipment: NI ELVIS II+, PC with LabVIEW, LDR (Photoresistor), Fixed Resistors (e.g., **10 kΩ**), Connecting Wires, Light Source (Flashlight/Smartphone), LED (with 220 Ω resistor).

Part 1: Theoretical Background

1.1 The Light Dependent Resistor (LDR)

An LDR, or photoresistor, is a passive electronic component whose resistance decreases as the intensity of incident light increases. It is typically made of a high-resistance semiconductor material, such as Cadmium Sulfide (CdS). When photons of light are absorbed by the material, they give bound electrons enough energy to jump into the conduction band. The resulting free electrons conduct electricity, thereby lowering the resistance.

1.2 The LDR Equation and Curve

The relationship between the LDR's resistance (R_{LDR}) and the light illuminance (E, *measured in Lux*) is non-linear and typically follows an inverse power law:

$$R_{LDR} = A \times E^{-\alpha} \quad (1)$$

- R_{LDR}= Resistance of the LDR in Ohms (Ω)
- E = Light Illuminance in Lux (lx)
- A = Constant depending on the specific LDR material and manufacturing (Resistance at 1 Lux)
- α = Constant determining the slope of the curve (*typically between 0.7 and 0.9*)

- Graphical plot example for $A = 500000$ and $\alpha = 1.2$

E-mail : gaga.ahmad@gmail.com

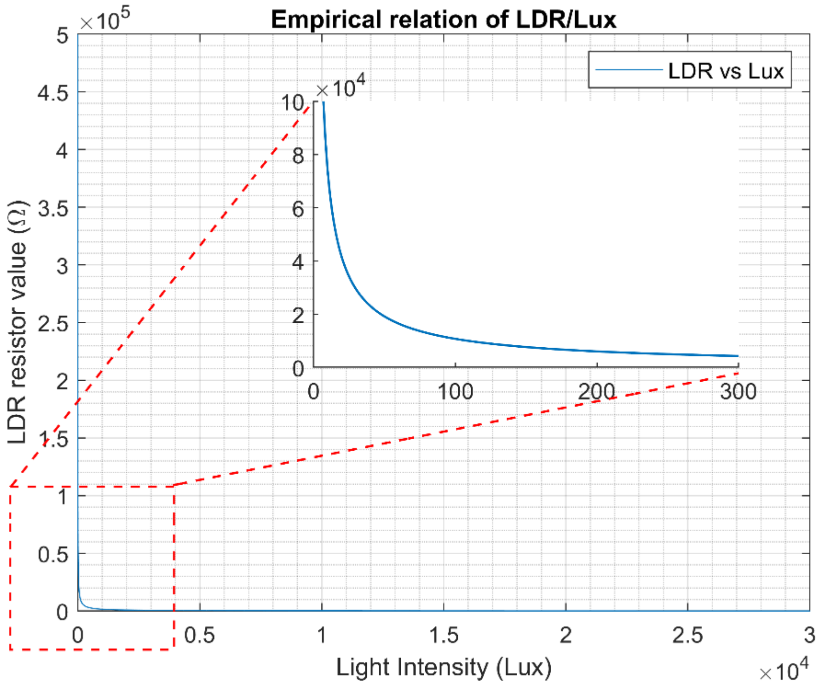

Figure 1: Graphical plot example for A=500000 and α=1.2

Because of this non-linear relationship, the characteristic curve is usually plotted on a **log-log scale**, which transforms the power-law curve into a straight line:

$$log(R_{LDR}) = log(A) - \alpha . log(E) \tag{2}$$

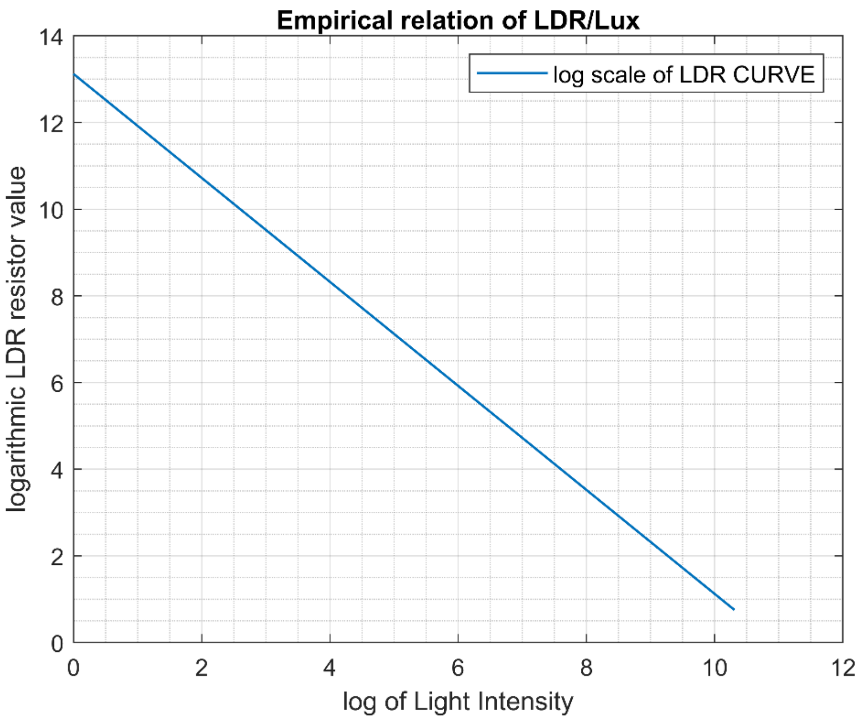

Figure 2: characteristic curve plotted on a log-log scale

1.3 Signal Conditioning (The Voltage Divider)

The NI ELVIS II+ Data Acquisition (DAQ) system measures voltage, not resistance directly. To acquire the LDR's data, we must convert its varying resistance into a varying voltage using a voltage divider circuit.

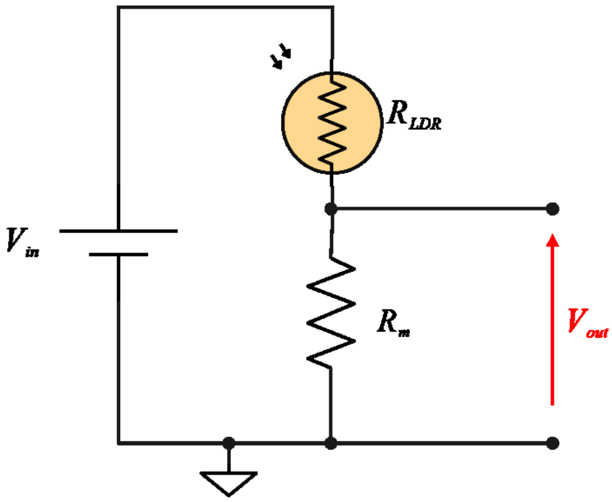

Figure 3: LDR voltage divider schematic

By placing the LDR in series with a fixed measuring resistor (R_m) as shown in Figure 3 and applying a known excitation voltage (V_{in}), the output voltage (V_{out}) measured across R_m is calculated as:

$$V_{out} = V_{in} \times \frac{R_m}{R_m + R_{LDR}} \tag{3}$$

By configuring the circuit this way, as light increases (R_{LDR} decreases), the measured voltage V_{out} will intuitively increase.

Part 2: Hardware Schematic & Setup (NI ELVIS II+)

- **Instructions for Prototyping Board (***at each step see Figure 1***):**

1. **Power Supply:** Connect the $+5V$ terminal on the NI ELVIS prototyping board to one leg of the LDR. ($V_{in} = +5\,Volts$).
2. **Voltage Divider:** Connect the other leg of the LDR to one leg of the fixed resistor R_m (*Use a* $\mathbf{10\ k\Omega}$ *resistor to start*).
3. **Ground:** Connect the remaining leg of R_m to the **Ground (GND)** terminal.
4. **Data Acquisition:** Connect the junction between the LDR and R_m to the Analog Input Channel 0 positive terminal (**AI 0+**). Connect the **(AI 0-)** terminal to Ground to complete a single-ended or pseudo-differential measurement.

Part 3: Step-by-Step Practical Tasks

Task 1: Static Characterization of the LDR (Manual Calibration)

Objective: *Determine the constants A and α for the specific LDR.*

1. Use the NI ELVIS Digital Multimeter (DMM) function to measure the LDR's resistance directly (disconnect it from the 5V supply first).
2. Use a smartphone light meter app (Lux meter) to measure the light intensity at various distances from a light source.
3. Record pairs of data: (E_{lux}, R_{LDR}) in an Excel table. Take at least five different points (*from total darkness to direct bright light*).
4. Plot $log(R)$ vs $log(E)$ and determine the equation of the line (*as shown in Figure 2*).
5. Calculate A and α.

Help:

- you can determine the constants A and α experimentally using two measured points and equation (1):

$$(E2, R_{LDR}1)\ ;\ (E2, R_{LDR}2)$$

$$R_{LDR1} = A \times E1^{-\alpha}$$

$$R_{LDR2} = A \times E2^{-\alpha}$$

$$\alpha = \frac{\ln(R_{LDR1}/R_{LDR2})}{\ln(E_2/E_1)}$$

- Equation 2 is a linear correlation between $log(E)$ and $log(R_{LDR})$, we suppose the following equivalence:

$$X = log(E)$$

$$Y = log(R_{LDR})$$

We obtain the following equation:

$$Y = b.X + c$$

With: $b = -\alpha = \frac{\Delta Y}{\Delta X}$

And: $c = log(A)$; With: $A = R_{LDR}(1or2) \times E(1or2)^{\alpha}$

Task 2: Real-Time Voltage Acquisition in LabVIEW

Objective: *Build the basic DAQ interface.*

1. Rebuild the voltage divider circuit on the ELVIS board.
2. Open LabVIEW and create a new Blank VI.
3. In the Block Diagram, drop a **DAQ Assistant** node. Configure it to acquire *Analog Input →Voltage* using the appropriate ELVIS $AI0$ channel. Set acquisition to *Continuous Samples* at a rate of 100 Hz (10 ms) inside a While Loop.
4. Wire the output to a **Waveform Chart** on the Front Panel.
5. Run the VI and cast a shadow over the LDR with your hand. Observe the voltage drops and spikes.

E-mail : gaga.ahmad@gmail.com

Task 3: Signal Processing (Voltage to Lux)

Objective: *Reconstruct the physical quantity.*

1. Using LabVIEW mathematical nodes (or a Formula Node), program the inverse of the voltage divider equation to isolate and calculate real-time resistance:

$$R_{LDR} = R_m \times \frac{V_{in} - V_{out}}{V_{out}} \tag{4}$$

2. Using the constants A and α found in Task 1, program the LabVIEW block diagram to calculate real-time Lux:

$$E = \left[\frac{R_{LDR}}{A}\right]^{-\frac{1}{\alpha}} \tag{5}$$

3. Display the calculated Lux on a graphical "Gauge" or "Meter" on the Front Panel.

Task 4: Supervision and Smart Control (Application)

Objective: *Use the acquired data to trigger a physical action, simulating a smart street lighting system.*

1. In LabVIEW, add a **Comparison Node** (e.g., *Less Than*).
2. Set a threshold value (e.g., 50 Lux). Compare your real-time Lux calculation to this threshold.
3. Wire the Boolean output of this comparison to a Front Panel LED indicator (Warning: Low Light).
4. **Hardware Loop:** Wire the same Boolean output to a **DAQ Assistant (Digital Output)**. Connect an actual LED (*with a* $\mathbf{220\ \Omega}$ *protective resistor*) to the corresponding Digital Out pin on the NI ELVIS board.
5. *Result:* When the student covers the LDR, the LabVIEW virtual LED will light up, and the physical LED on the ELVIS board will turn on simultaneously.

ANNEXE: *Typical practical values for standard LDRs*

Light Condition	*Typical Illuminance (Lux)*	*Typical LDR Resistance*
Complete darkness	$0\ lux$	$(1M\Omega)$ to $(20M\Omega)$
Moonlight	$0.1 - 1\ lux$	$(500k\Omega)$ to $(5M\Omega)$
Dim room	$5 - 20\ lux$	$(100k\Omega)$ to (500Ω)
Normal indoor lighting	$100 - 500\ lux$	$(10k\Omega)$ to $(100k\Omega)$
Classroom / office	$300 - 1000\ lux$	$(2k\Omega)$ to $(20k\Omega)$
Outdoor cloudy daylight	$1000 - 10000\ lux$	(500Ω) to $(5k\Omega)$
Direct sunlight	$30000 - 100000\ lux$	(100Ω) to $(1k\Omega)$

E-mail : gaga.ahmad@gmail.com